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Isolation and structural characterization of a novel polysaccharide from *Hericium erinaceus* fruiting bodies and its arrest of cell cycle at *S*-phage in colon cancer cells

Jing-Yu Liu [a,b,c], Xiao-Xiao Hou [b], Zhuo-Yu Li [a,c,*], Shu-Hua Shan [a,c], Ming-Chang Chang [b,c], Cui-Ping Feng [b], Yin Wei [b]

[a] Key Laboratory of Chemical Biology and Molecular Engineering of National Ministry of Education, Institute of Biotechnology, Shanxi University, Taiyuan, Shanxi 030006, China
[b] College of Food Science and Engineering, Shanxi Agricultural University, Taigu, Shanxi 030801, China
[c] Collaborative Innovation Center of Quality and Efficiency of Loess Plateau Edible Fungi, Taigu, Shanxi 030801, China

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ABSTRACT

The fruiting body of *Hericium erinaceus* has been used to treat digestive system disorder-related diseases for over 2000 years in China. A novel polysaccharide, HEFP-2b, was obtained from *H. erinaceus* fruiting bodies. Physical and chemical analysis showed that HEFP-2b consisted of fucose, galactose, glucose, and mannose in molar ratio of 11.81:22.82:44.28:21.09, and that its molecular weight was 3.252×10^4 Da. The backbone of HEFP-2b consisted of →6)-linked-α-D-Glc*p*-(1→ and →4)-β-D-Gal*p*-(1→ and →3,6) -α-D-Man*p* linkage, with two side-branching units of (1→ and →6)-β-D-Gal*p* and (1→ and →4)-α-D-Man*p*, terminated by Glc and Fuc. The results of the 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT) assay and cell cycle arrest experiments revealed that HEFP-2b considerably inhibited the growth of colon cancer cells (HCT-116) *in vitro*. The growth inhibitory effects of HEFP-2b correlated with their ability to arrest the cell cycle at the *S*-phase. Our results will provide valuable information for future studies on HEFP-2b as a novel health-promoting functional food ingredient that can be used for treating colon cancer.

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1. Introduction

Polysaccharides have been use as natural biomaterials and functional food and health care, and exhibit diverse roles *in vivo*, such as immuno-regulatory, anti-oxidation, anti-inflammatory, and anti-tumor functions [1,2]. Recently, an increasing number of functional mushroom-derived polysaccharides have been identified and investigated as biological response modifiers and effective nonspecific agents for cancer-associated therapy and as functional food [3]. Mushroom-derived polysaccharides have been considered a critical source of powerful immune-modulators in cancer therapy [4].

Hericium erinaceus, commonly known as Houtou, Lion's Mane, or Yamabushitake, is as a medicinal mushroom with well-known health-promoting effects [5]. Its fruiting bodies have been used to treat digestive system disorder-related diseases in China for over 2000 years [6]. Some biologically active ingredients extracted from *H. erinaceus*, such as terpenoids, polyphenols, cerebrosides, or polysaccharides, have been recently reported to possess various biological activities, including anti-oxidative, immuno-modulatory, hypolipidemic, anti-chronic atrophic gastritis, and anti-cancer activities [7–10]. Polysaccharides derived from *H. erinaceus* (HEPs) have been used as potential drugs for the treatment of liver and gastric cancers, and several reports have indicated that these polysaccharides promote the apoptosis of gastric cancer and liver cancer cells [11,12]. Recently, HEPs have attracted attention due to their potential application in functional food and biomedical products [13,14]. However, up till now, information regarding the effects of HEPs on anti-colon cancer activities is limited.

In the present study, a novel polysaccharide (HEFP-2b) was extracted and purified from *H. erinaceus* fruiting bodies. The chemical and structural properties of HEFP-2b were sequentially investigated. The data of cell viability test and molecular studies showed that the growth inhibitory effects of HEFP-2b correlated with their ability to arrest the cell cycle at the S-phase. The results of this study might provide valuable information for future studies on HEFP-2b as a potential functional food ingredient or pharmaceutical agent for treating colon cancer.

* Corresponding author at: Key Laboratory of Chemical Biology and Molecular Engineering of National Ministry of Education, Institute of Biotechnology, Shanxi University, Taiyuan, Shanxi 030006, China.
E-mail address: lzy@sxu.edu.cn (Z.-Y. Li).

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2. Materials and methods

2.1. Materials and reagents

Fungal materials: The fruiting bodies of *H. erinaceus* were provided by the Shanxi Engineering Research Center of Edible Fungi (Shanxi, China). A homogeneous powder of H. erinaceus fruiting bodies was obtained by processing fresh fruiting bodies, which were dried (50 °C, 72 h), crushed, and passed through a 100-mesh sieve.

Reagents: DEAE-cellulose-52 and Sephacryl S-400 were purchased from Solarbio (Beijing, China) and GE Healthcare (Chicago, IL, USA). Monosaccharides, including D-glucose, D-galactose, D-mannose, D-fucose, D-rhamnose, D-ribose, D-fructose, D-arabinose, and D-xylose were purchased from Merck Corp (Darmstadt, Germany).

3-(4,5-Dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide (MTT), dimethyl sulfoxide (DMSO), bovine serum albumin (BSA), phosphate-buffered saline (PBS) and Tris-buffered saline-Tween 20 (TBST) were purchased from Sigma-Aldrich (St. Louis, MO, USA). In addition, chromatographic grade trifluoroacetic acid (TFA) was purchased from Aladdin (Shanghai, China). Antibodies against cyclin A2, CDK1, CDK2, $P57^{kip2}$, and MCM5 were purchased from Bioworld (St Louis, MO, USA) and Beyotime Institute of Biotechnology (Haimen, China). GAPDH was purchased from Abmart (Shanghai, China).

2.2. Extraction and purification of HEFP-2b

Approximately 20 g powder of dried *H. erinaceus* fruiting bodies and 800 mL distilled ultrapure water (Milli-Q ® Reference, Merck, GER) were placed in a 1000 mL beaker, and the mixture was extracted for 3 h at 80 °C in a constant-temperature water bath. The supernatant was obtained *via* centrifugation (Beckman, Germany) at 4500 ×*g* for 20 min and was concentrated using reduced pressure in an RV-10 rotary evaporator (IKA, China). A hybrid solution, which was composed of 10.6% potassium ferricyanide solution and 21.9% zinc acetate solution, was obtained, and the concentrate was then mixed at a 1:1:10 ratio based on published protocols [15]. After mixing well, the mixture was allowed to rest for 30 min, and was subsequently centrifuged at 3000 ×*g* for 20 min. The supernatant was concentrated and dialyzed against ultrapure water (molecular weight cutoff of 3500 Da) for 48 h to remove small molecular components. Thereafter, the precipitate was obtained after injecting anhydrous ethanol (the final alcohol concentration in mixed liquor reached 80%) for 12 h at 4 °C and lyophilizing the crude polysaccharides from *H. erinaceus* fruiting bodies (HEFPs).

The crude polysaccharides were further separated and purified following a previously reported method with minor modifications [10]. Briefly, the obtained HEFPs were dissolved in distilled ultrapure water to a concentration of 10 mg/mL, loaded on a DEAE-cellulose-52 ion exchange column (2.8 cm × 32 cm), and sequentially eluted with distilled ultrapure water and 0.1 mol/L NaCl solution at the flow rate of 2 mL/min. The eluted components, were collected and reloaded into a gel permeation chromatography column of Sephacryl S-400 (XK26 × 100 cm), and then eluted with 0.2 mol/L NH_4HCO_3 solution at a flow rate of 3 mL/min. The eluent was collected and detected at 490 nm absorbance using the phenol-sulfuric acid method. Eluents with the same elution curve peak were combined using an auto collector, and dialyzed against with ultrapure water for 48 h (3500 Da). Finally, the polysaccharide fraction was lyophilized and named HEFP-2b for subsequent studies. The protein content of the polysaccharides was determined using Coomassie Brilliant Blue.

2.3. Structural characterization of HEFP-2b

2.3.1. Determination of molecular weight

The molecular weight and the radius of gyration of HEFP-2b were measured on size exclusion chromatography (SEC) columns (Wyatt Technology, USA) coupled with a multi-angle laser light scattering (MALLS) system (DAWN HELLEOS, Wyatt Technology, USA, equipped with a He—Ne laser, $\lambda = 663.7$ nm) and a refractive index (RI) detector (Optilab T-rEX, $\lambda = 658.0$ nm) [16]. The polysaccharide sample was dissolved in ultrapure water at 25 °C and then filtered through a 0.22 μm Millipore filter prior to injection into the SEC-MALLS-RI system. The flow rate was set at 0.5 mL/min, and the injection volume and concentration were 100 μL and 3 mg/mL, respectively. The values of dn/dc were determined using a RI detector (Optilab T-rEX, Wyatt Technology, USA) at 658.0 nm and 25 °C. The data acquisition and analysis were performed using the Astra software (Version 6.1, Wyatt Technology, USA).

2.3.2. Analysis of monosaccharide composition

The HEFP-2b (20 mg) sample was hydrolyzed using 2 mol/L TFA (4 mL) at 120 °C for 6 h and dried using a AG-DCY-24G nitrogen blowing instrument (Bingyue Electronic, China) after washing thrice with 1.5 mL methanol [17]. The monosaccharide composition of HEFP-2b was measured using Dionex ICS3000 ion chromatography (Thermo Scientific, USA) with Carbopac PA20 (2 × 250 mm) (Dionex, USA). Nine standard monosaccharide sugars (D-glucose, D-galactose, D-mannose, D-fucose, D-rhamnose, D-ribose, D-fructose, D-arabinose, and D-xylose) were mixed for use as standards. Approximately 25 μL hydrolyzed sample solutions, which were previously filtered using 0.22 μm hydrophilic membranes, were injected into the sample system for ion-exchange chromatography in each run. The column was eluted with a mixed eluent consisting of 95% ultrapure water and 5% 100 mmol/L NaOH solution at the flow rate of 1 mL/min. The temperature of the column was maintained at 30 °C and the signal was obtained using a by conductance detector.

2.3.3. UV and FT-IR spectroscopy analysis

The HEFP-2b sample was prepared into 1 mg/mL aqueous solution, and the ultraviolet-visible (UV) absorption spectra of the prepared sample was measured in the range of 190–800 nm using a Cary60 ultraviolet-visible spectrophotometer (Agilent, USA) with a 1 nm sweep spacing. FT-

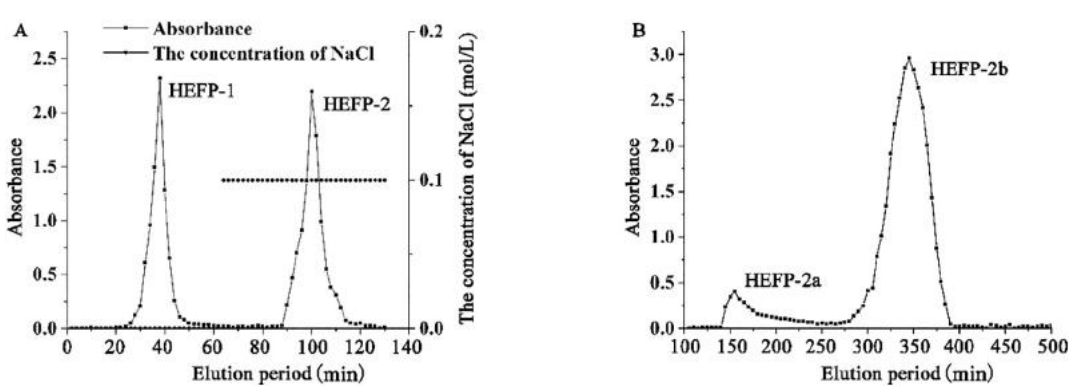

Fig. 1. Elution profiles of DEAE-52 and Sephacryl-S400 of HEFPs. (A) Elution profiles of DEAE-52 of the HEFPs; (B) Elution profiles of Sephacryl-S400 of the HEFP-2.

Table 1
Average molecular weight, dispersion coefficient and molecular radius of HEFP-2b.

Sample	M_w (Da)	M_n (Da)	M_z (Da)	M_w/M_n	Rg (nm)	k	α
HEFP-2b	3.252×10^4	1.833×10^4	1.258×10^4	1.775	44.0	0.04	0.57

Note: M_w: weight average molecular weight; M_n: number average molecular weight; M_z: z-average molecular weight; R_g: radius of gyration; k: the intercept at the Y axis; α: the critical slope values.

IR analysis of HEFP-2b was performed using a TENSOR 27 Fourier transformed infrared spectrophotometer (FT-IR, Bruker, Germany), and the samples were analyzed using the KBr tableting method [18]. Briefly, the freeze-dried HEFP-2b sample (1 mg) was briefly mixed to uniformity with dried KBr powder (150 mg) and subsequently measured in the range of 400–4000 cm^{-1} after pressing into pellets (1 mm) using the FW-4A powder press machine (Tuopu, China).

2.3.4. Methylation analysis

The freeze-dried polysaccharide was methylated using Chen's method with some modifications [19]. The HEFP-2b sample (1 mg) was dissolved in 0.5 mL NaOH-DMSO with heating at 30 °C for 30 min, and then methylated using 0.1 mL iodine methane in the dark for 10 h. This methylation procedure was repeated thrice and terminated by the addition of 1 mL distilled water. The methylated polysaccharide was further extracted thrice with 0.5 mL dichloromethane for three times, and the dichloromethane extract was evaporated to dryness at 30 °C. The dried methylated polysaccharide was dissolved in 0.1 mL trifluoroacetic acid at 121 °C for 1.5 h and then the solution was evaporated to dryness at 30 °C. Ammonium hydroxide (0.05 mL) and 0.05 mL sodium borodeuteride were sequentially added and incubated at 30 °C for 2.5 h. The reaction was terminated by adding 0.02 mL acetic anhydride, followed by drying of the extract under nitrogen. Acetic anhydride (0.25 mL) was added and incubated at 100 °C for 2.5 h. Acetylation was performed according to the above-mentioned method. The methylation of the polysaccharide was confirmed based on the disappearance of the OH band at 3200–3700 cm^{-1} in the IR spectrum [20] and further analyzed using gas chromatography coupled with mass spectrometry (GC–MS) (Agilent 6890A-5975C, Agilent Technologies Inc. CA, USA) using a BPX70 capillary column with a temperature gradient of 140–230 °C at 3 °C min^{-1} and maintenance at 230 °C for 20 min.

2.3.5. NMR spectroscopy analysis

The freeze-dried HEFP-2b sample (20 mg) was dissolved in 5 mL deuterium oxide (D_2O) before nuclear magnetic resonance (NMR) analysis. The NMR experiment was performed using standard Bruker software (Bruker, USA), and the ^1H and ^{13}C spectra were obtained on a 600 MHz AVIII HD NMR spectrometer (Bruker, Germany) and recorded at 25 °C.

Table 2
GC–MS data of methylation analysis of HEFP-2b.

Methylated sugar	Retention time (min)	Mass fragments (m/z)	Linkage	Molar ratio
2,3,4-Me3-Fuc	7.65	59, 72, 89, 118,175	T-	5.26
2,3,4,6-Me4-Glc	9.71	87, 102, 118, 145, 162, 205	T-	19.42
2,3,4-Me3-Glc	14.69	71, 102, 118, 162, 189, 233	1,6-	33.15
2,3,6-Me3-Gal	14.99	71, 118, 162, 203, 233	1,4-	13.83
2,3,4-Me3-Gal	16.43	71, 102, 118, 162, 233	1,6-	8.23
2,6-Me2-Man	18.75	59, 87, 118, 160, 189, 234, 305	1,3,6-	11.55
2,4,6-Me3-Man	13.07	71, 101, 118, 161, 203, 234, 277	1,3-	8.55

2.3.6. Ultrastructural analysis

The freeze-dried HEFP-2b sample was applied onto a copper metal stub, and a layer of the conductive film was sprayed over the sample with a vacuum sputter coater [21]. The morphological properties of the HEFP-2b sample were observed using a scanning electron microscope (JEOL, JSM-6490LV, Japan).

2.4. Effects of HEFP-2b on inhibition of HCT-116 cells

2.4.1. Cell culturing

FHC, HCT-116 cells, which were obtained from cancerous human colorectal tissues, were provided by the China Center for Culture Collection (Wuhan, China). FHC (control) and HCT-116 cells were grown in Ham's F12 medium containing 10% (v/v) fetal bovine serum (FBS). Cultures were maintained at 37 °C with 5% CO_2 in a humidified incubator (Thermo Fisher, USA) for 24 h. The freeze-dried HEFP-2b sample was dissolved serially in the corresponding media and adjusted to varying concentrations after filtering with a 0.22 μm filter membrane.

2.4.2. Cell viability assay

The ability of HEFP-2b to inhibit the viability of the FHC (control) and HCT-116 cells was investigated using the MTT assay. Cells were spread onto a 96-well plate at a density of 6×10^3 cells per well and were cultured overnight at 37 °C. The media in the wells was discarded, and subsequently 100 μL of fresh medium containing different concentration of HEFP-2b (0, 0.2, 0.4, 0.6, 0.8 and 1.0 mg/mL) were added to each well; five replicates were made for each concentration well. The treatments were maintained for 48 h. Then, 20 μL MTT solution (5 mg/mL) was added to each well and the plate was incubated for 4 h. Finally, the supernatant was removed, 150 μL of DMSO was added to each well to dissolve any purple crystals (Formazan), and the

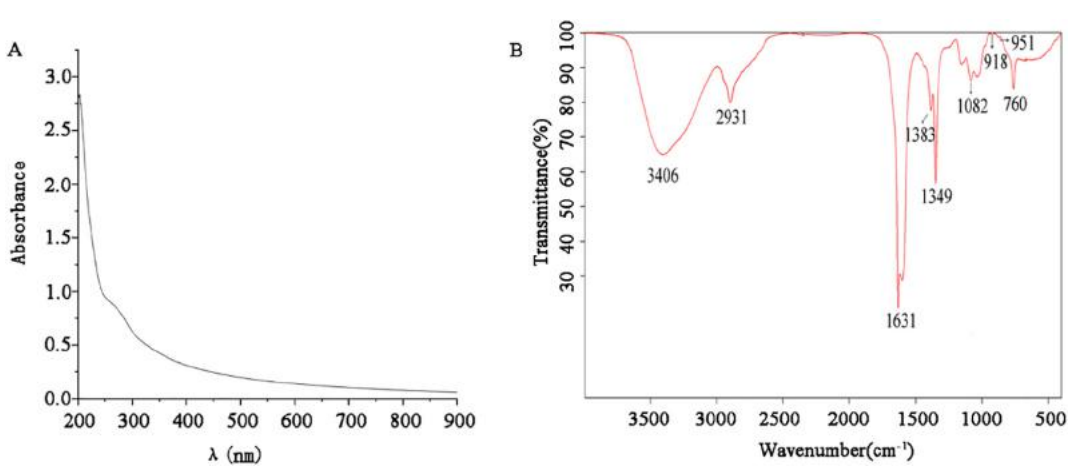

Fig. 2. The ultraviolet and FT-IR spectra of HEFP-2b. (A) The ultraviolet spectrum. (B) The FT-IR spectrum.

absorbance of each well was measured at 570 nm using an Infinite F50 microplate reader (Tecan, Switzerland).

2.4.3. Cell cycle analysis

HCT-116 cells at logarithmic growth phase were seeded in a 6-well cell culture plate with 1×10^6 cells per well and incubated for 12 h. Then, the fresh medium containing HEFP-2b was added and incubated for 48 h. The cells were trypsinized, centrifuged, fixed with 70% ethanol, and stored at 4 °C for 12 h. Cells were washed twice with PBS and then re-suspended with staining buffer (Cell Cycle Analysis Kit, Beyotime, China) containing 500 μL PI, 100 μL RNase A, and 1 mL PBS. The cell cycle was detected on an Accuri C6 flow cytometry (BD Biosciences, USA).

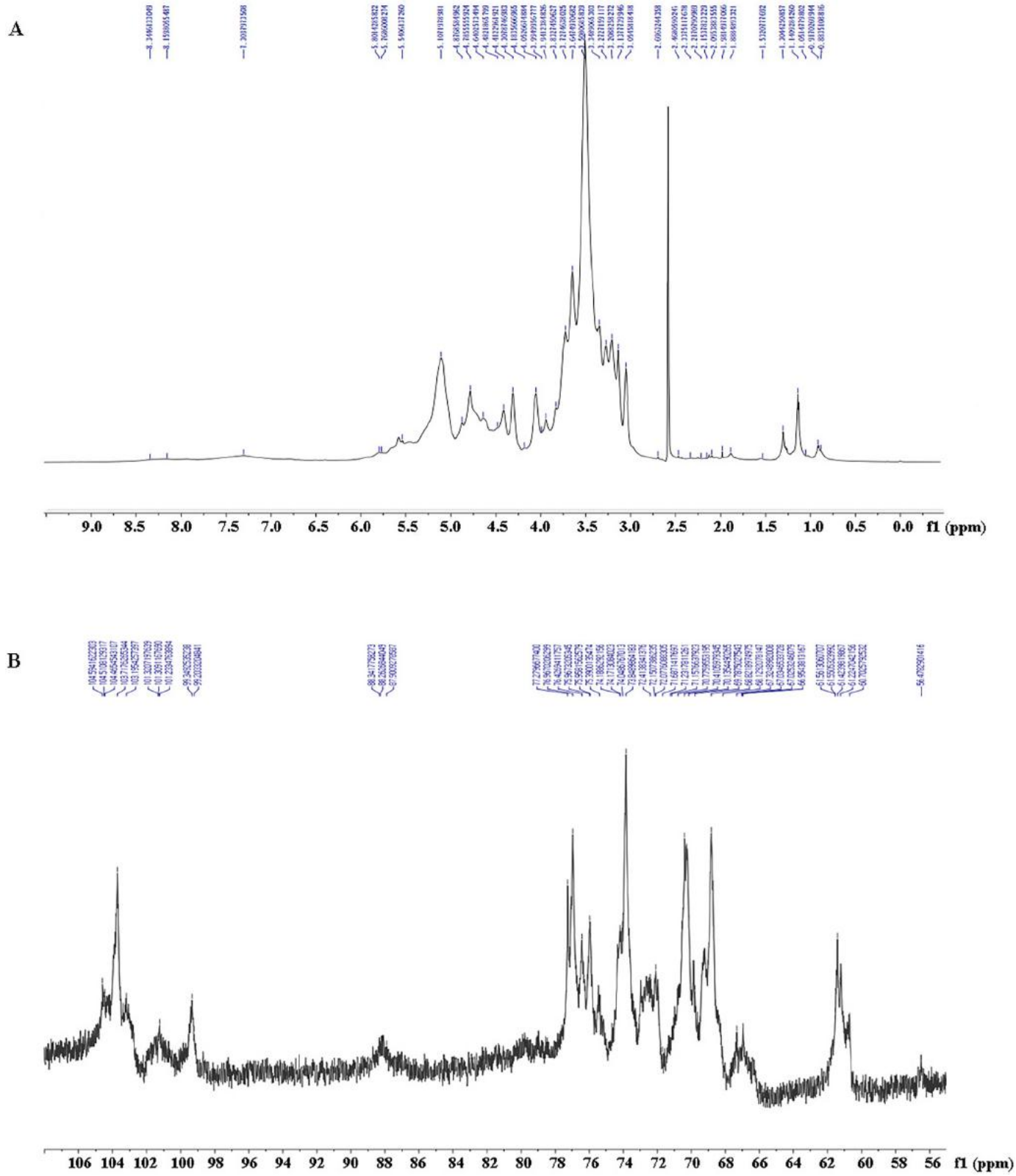

Fig. 3. NMR spectra of HEFP-2b. (A) ^{1}H NMR spectrum; (B) ^{13}C NMR spectrum.

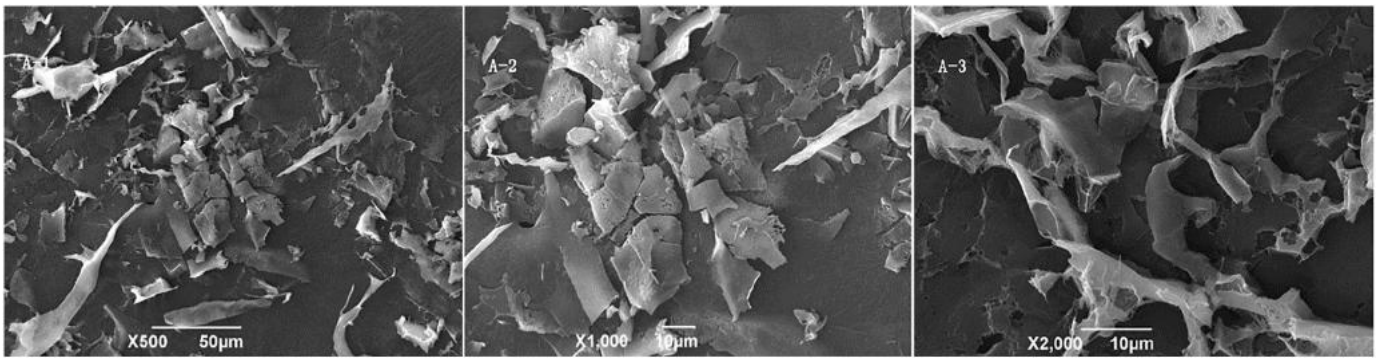

Fig. 4. SEM images of HEFP-2b. "A-1" means magnification 500×, bar indicates 50 μm; "A-2" means magnification 1000×, bar indicates 10 μm; "A-3" means magnification 2000×, bar indicates 10 μm.

2.4.4. Western blotting

Cells were trypsinized, collected, and lysed using cell lysis buffer on ice for 40 min after washing thrice with PBS. The protein concentration was determined using a bicinchoninic acid (BCA) protein assay kit (Beyotime, China). Sixty micrograms protein was loaded in each lane and was separated using 10% sodium dodecyl sulfate-polyacrylamide gel electrophoresis (SDS-PAGE) and transferred to a polyvinylidene fluoride (PVDF) membrane. After the transfer, the PVDF membrane was blocked with 5% skim milk for 4 h and was incubated overnight with the primary antibody. After washing with TBST, the membrane was incubated with secondary antibodies at room temperature for 2 h. The labeled proteins of interest were visualized by exposing the blots to an X-ray film in a dark room using an enhanced chemiluminescence (ECL) kit (Sigma, USA).

2.5. Statistical analyses

All tests were performed at least thrice. The results were presented as means ± standard deviations (S ± D). The IC50 values were determined using regression analysis of the SPSS software (IBM, USA), followed by the Student's *t*-test. The levels of statistical significance were accepted at P-value < 0.05.

3. Results and discussion

3.1. Extraction and purification of HEFP-2b

In the present study, the polysaccharides (HEFPs) were extracted from *H. erinaceus* fruiting bodies using a novel extraction method involving hot water coupled with ferrocyanide-zinc acetate treatment [15,23]. The polysaccharide and protein contents of freeze-dried HEFPs were determined to be 76.16 ± 2.74% and 2.41 ± 0.56%, respectively. The freeze-dried HEFPs were then sequentially eluted into two components named HEFP-1 and HEFP-2 using a DEAE-cellulose-52 ion exchange column and distilled ultrapure water and 0.1 mol/L NaCl solution, respectively (Fig. 1A). The polysaccharide contents of HEFP-1 and HEFP-2 were 83.26 ± 1.69% and 86.94 ± 1.27%, respectively. Furthermore, the protein contents of HEFP-1 and HEFP-2 were 1.16 ± 0.25% and 2.78 ± 0.42%, respectively. According to the results of the MTT assay of the two components (unpublished), the HEFP-2 sample was further eluted though a gel permeation chromatography column (Sephacryl S-400) using 0.2 mol/L NH_4HCO_3 solution. Two fractions, HEFP-2a and HEFP-2b, were obtained (Fig. 1B). The yield of the HEFP-2b fraction was the highest (71.8%), whereas that of HEFP-2a was considerably low (10.6%). Next, MTT assay test was used to determine the effect of HEFP-2a and HEFP-2b on cell viability; results showed that HEFP-2b inhibited HCT-116 cells more effectively than HEFP-2a. Therefore, only HEFP-2b was analyzed for subsequent structural characterization and anti-tumor activities. The polysaccharide content of HEFP-2b was determined to be 95.24 ± 1.81% and the protein content was reduced to 0.84 ± 0.29%.

3.2. Molecular weights of HEFP-2b

As shown in Table 1, the average molecular weight (M_w), number average molecular weight (M_n) and z-average molecular weight (M_z) of HEFP-2b were determined to be 3.252×10^4 Da, 1.833×10^4 Da and 1.258×10^4 Da, respectively, using the SEC-MALLS analysis system (Wyatt Technology, USA). The results showed that the average molecular weight of HEFP-2b was higher than those of the previously reported polysaccharides from *H. erinaceus* [8].

The coefficient of M_w/M_n (1.775) was >1 and close to 2. This indicated that the molecular weight distribution of the HEFP-2b sample was narrower and that the polymer structure was more uniform [24]. Furthermore, the chain conformation of the HEFP-2b sample in aqueous solution was identified using the following formula: $R_g = kM_w^{\alpha}$ (R_g: radius of gyration; k: the intercept at the Y axis s; α: the critical slope value) [25]. The result showed that the calculated R_g of HEFP-2b (0.44 nm) was >0.33 nm and close to 0.5 nm. Hence, we speculated that the HEFP-2b sample consisted mostly of relatively stretched

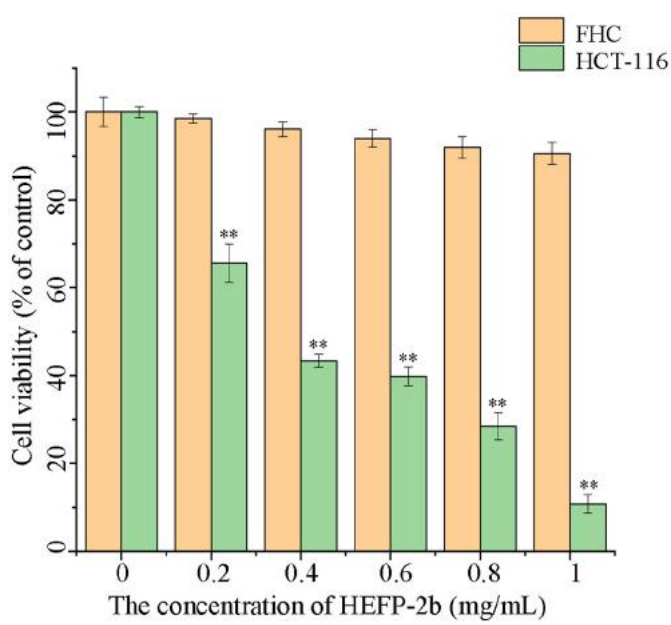

Fig. 5. Effects of HEFP-2b on the growth of FHC and HCT-116 cells lines *in vitro*. Data represent means ± SD (n = 3). *p < 0.05, ** p < 0.01 compared with untreated control cells.

molecular chains, which might have contributed to high molecular hydrophilicity and excellent molecular solubility in aqueous solution [26].

3.3. Monosaccharide composition of HEFP-2b

The monosaccharide composition of HEFP-2b was determined using Dionex ICS3000 ion chromatography with Carbopac PA20 (Dionex, USA). Based on the elution curve of monosaccharide standards, HEFP-2b was speculated to mainly consist of fucose (Fuc), galactose (Gal), glucose (Glc) and mannose (Man) monohydrates at a molar ratio of 11.81:22.82:44.28:21.09 (Fig. S1). Furthermore, some arabinose (Ara), rhamnose (Rha), ribose (Rib), fructose (Fru) and xylose (Xyl) were also detected in HEFP-2b. The observation that the Gal (22.82%) content of HEFP-2b was high is somewhat different from those of the previously reported polysaccharides from *H. erinaceus* fruiting bodies, which were mainly composed of Glc, Man, and Fuc [14,27,28]. This might be due to differences in strains or extraction methods [29,30].

3.4. UV and FT-IR spectroscopy of HEFP-2b

The ultraviolet absorption spectrum of HEFP-2b (Fig. 2A) showed absence of no significant absorption peaks at both 260 nm and 280 nm. This indicated that HEFP-2b contained negligible amounts of proteins and nucleic acids.

As shown in Fig. 2B, the FT-IR spectra of HEFP-2b exhibited major typical absorption peaks of most polysaccharides. According to published literature [31–33], the intense characteristic peak at 3406 cm^{-1} represented the stretching of the O—H bond, and the weak absorption peak at 2931 cm^{-1} was due to the stretching vibration of C—H. The strong absorption peak at 1631 cm^{-1} was due to an amide carbonyl group, which is due to the stretching vibration of C═O in the -CO- group. The strong characteristic absorption peaks between 1383 cm^{-1} and 1349 cm^{-1} indicated the deformation vibration of the C—H bond. The strong absorption peak at around 1082 cm^{-1} indicated the presence of a pyranose ring, which corresponded to the absorption peaks of C-O-C and C—O single bonds. This result was consistent with the data of monosaccharide composition analysis. The characteristic absorptions at 934 cm^{-1} and 851 cm^{-1} indicated that the presence of β- and α-configurations [34].

3.5. Methylation analysis of HEFP-2b

To obtain further structural information, the HEFP-2b sample was methylated and analyzed using GC–MS. On the basis of the spectral database (the Complex Carbohydrate Research Center), seven derivatives were identified as 1,5-di-*O*-acetyl-6-deoxy-2,3,4-tri-*O*-methyl fucitol, 1,5-di-*O*-acetyl-2,3,4,6-tetra-*O*-methyl glucitol, 1,5,6-tri-*O*-acetyl-2,3,4-tri-*O*-methyl glucitol, 1,4,5-tri-*O*-acetyl-2,3,6-tri-*O*-methyl galactitol, 1,5,6-tri-*O*-acetyl-2,3,4-tri-*O*-methyl galactitol, 1,3,4,5-tetra-*O*-acetyl-

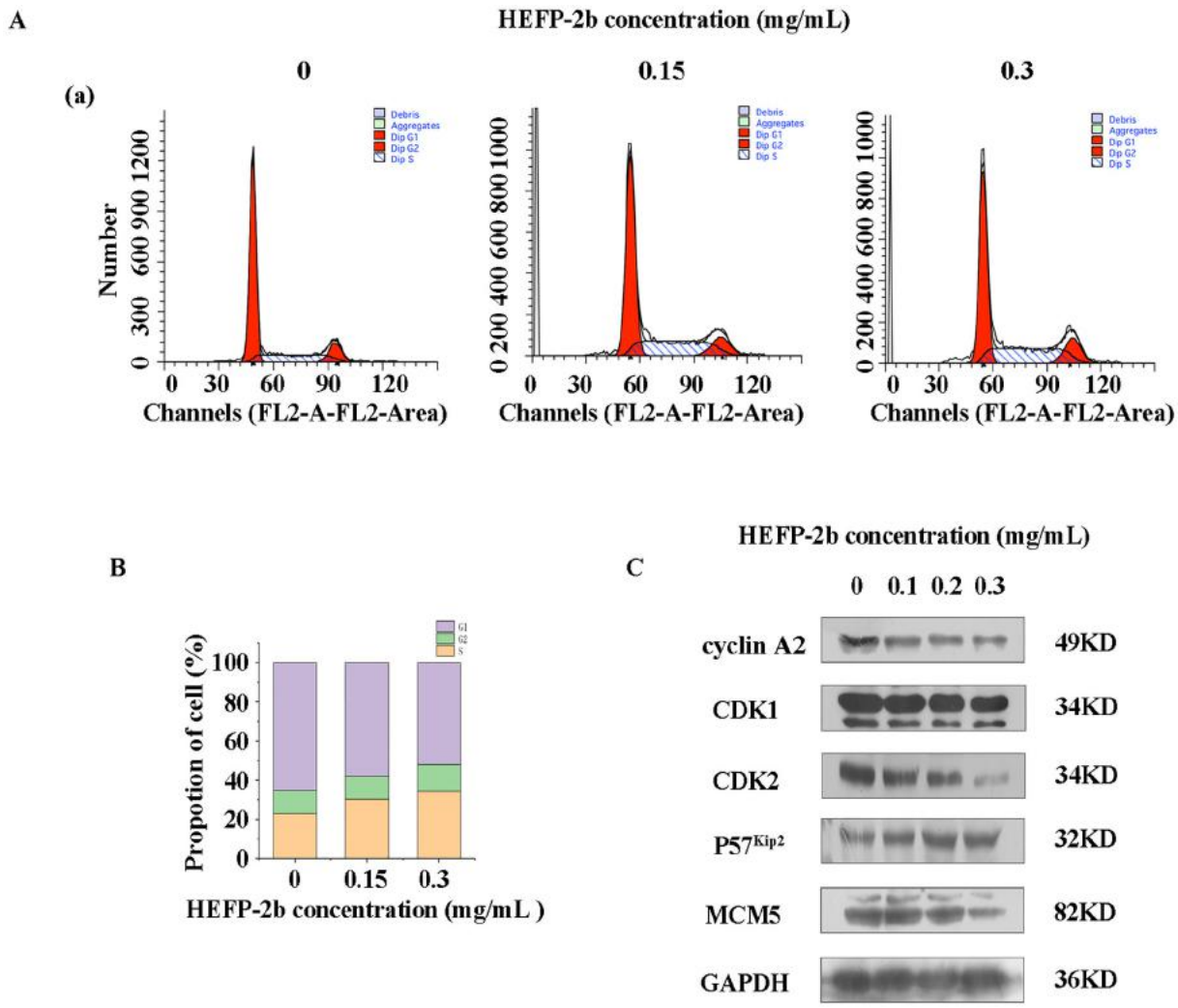

Fig. 6. Effect of HEFP-2b on cell cycle distribution in HCT-116 cells. (A) The cycle distribution of colon cancer cell after HEFP-2b treatment was detected by flow cytometry; (B) Quantitative analysis of cell cycle distribution; (C) Cell cycle relative protein expression profile was analyzed by western blotting and GAPDH was used as the reference.

2,6-di-*O*-methyl mannitol, and 1,3,5-tri-*O*-acetyl-2,4,6-tri-*O*-methyl mannitol with molar ratios of 5.26: 19.42: 33.15: 13.83: 8.23: 11.55: 8.55, respectively (Table 2). These results indicated that HEFP-2b was mainly composed of (1 → 6)-linked and terminal Glc (Residue A), (1 → 4) and (1 → 6)-linked Gal (Residue B), (1 → 3) and (1 → 6)-linked Man (Residue C), and terminal Glc and Fuc (Residue D).

3.6. NMR spectroscopy of HEFP-2b

The structural characteristics of HEFP-2b were investigated using one-dimensional ^{1}H NMR (Fig. 3A) and ^{13}C NMR (Fig. 3B). The result the presence of resonance signals of anomeric proton at δ_H 4.4–5.3 ppm in the ^{1}H spectrum (Fig. 3A) and anomeric carbon at δ_C 90–110 ppm in the ^{13}C spectrum (Fig. 3B). This indicated that HEFP-2b contained both α- and β- configuration in its structure [35,36], which was consistent with the results of the FT-IR spectrum. As shown in Fig. 3B, four monosaccharides (Fuc, Gal, Glc, and Man) mainly adopted the α-configuration as reported previously [37,38]. Briefly, the signals at δ_C 67.02, 99.20, 70.78, 71.23, 69.79, and 67.03 ppm were attributed to the C-1, C-2, C-3, C-4, C-5, and C-6 of α-D-Fuc, respectively. The signals at δ_C 104.51, 71.70, 73.85, 69.79, 75.96, and 61.56 ppm corresponded to C-1, C-2, C-3, C-4, C-5, and C-6 of β-D-Gal, respectively. The signals at δ_C 99.34, 72.15, 74.17, 70.78, 72.41, and 61.55 ppm were assigned to the C-1, C-2, C-3, C-4, C-5, and C-6 of α-D-Glc. The signals at δ_C 101.31, 71.23, 72.15, 68.13, 73.85, and 61.55 were assigned to the C-1, C-2, C-3, C-4, C-5, and C-6 of α-D-Man, respectively [39].

Based on the results of GC–MS and NMR, we concluded that the backbone of HEFP-2b might mainly consist of (1 → 6)-linked-α-D-glucose residues, (1 → 4)-linked-β-D-galactose residues, (1 → 3,6)-linked-α-D-mannose residues, and terminal glucose and fucose. Two branches consisted of (1 → 6)-linked-β-D-galactose and (1 → 3)-linked-α-D-mannose residues.

3.7. SEM of HEFP-2b

The apparent morphology of HEFP-2b was determined using scanning electron microscopy (SEM) (Fig. 4). HEFP-2b had a flat smooth shape and the particles were lamellar. The surface structure of HEFP-2b was similar to that of the heteropolysaccharide from *Murus alba* fruits [20].

3.8. Effects of HEFP-2b on growth of HCT-116 cells

To test the potential anti-colon cancer activity of HEFP-2b, MTT assays were performed to determine the effects of HEFP-2b on the proliferation of FHC (control), and HCT-116 cell lines *in vitro*. As shown in Fig. 5, HEFP-2b was able to effectively inhibit the growth of human colorectal carcinoma cell lines (HCT-116) in a dose-dependent manner, with IC50 values of 0.345 ± 0.026 mg/mL. HEFP-2b showed no obvious effect on the normal colonic epithelial cell line (FHC) at the same concentration. These results indicated that HEFP-2b considerably inhibited the proliferation of colon cancer cells *in vitro*, and that the cytotoxicity of HEFP-2b was negligible within the concentration range.

3.9. Analysis of HEFP-2b induced cell cycle arrest

Cell cycle arrest experiments were performed to further analyze the inhibitory effects of HEFP-2b on the growth of colon cancer cells. Results of flow cytometry analysis revealed S-phase arrests in HCT-116 cells after 48 h of HEFP-2b treatment (Fig. 6A). The result of quantitative analysis further indicated that the proportion of S-phase cells increased gradually with increase in HEFP-2b concentration (Fig. 6B).

It is well-known that many anti-cancer agents can arrest the cell cycle at specific phases. To investigate how HEFP-2b induced cell cycle arrest, western blotting analysis on cyclin A2, cyclin-dependent kinases (CDK1 and CDK2), and CDK inhibitor ($P57^{kip2}$) was performed in HCT-116 cells treated with an increasing concentration of HEFP-2b. S-phase arrest of the cell cycle is known to be controlled by cyclins, CDKs, and CDK inhibitors (CDKIs) [41,42]. Cyclin A2 and CDKs are essentials for driving the transition between the different phases of the cell cycle in colon cancer [43,44]. $P57^{kip2}$, a CDKI, is considered an attractive target for potential anticancer agents in clinically therapeutic interventions against different cancers [45,46]. As shown in Fig. 6C, CDK1, CDK2 and their regulatory partner protein cyclin A2 were significantly down-regulated in HCT-116 cells treated with increasing concentrations of HEFP-2b. Consistent with this, the expression of P57kip2 and the cyclin A2-CDK2 complex inhibitor was remarkably increased in the cells.

To confirm this, the changes in the expression of a mini-chromosomal maintenance protein (MCM5) were further investigated. MCM proteins are essential for regulating the initiation of DNA replication in the S-phase, and increase in MCM5 expression has been shown to be associated with tumor proliferation in colon cancer [47,48]. As expected, MCM5 expression significantly decreased in cells treated with an increasing concentration of HEFP-2b (Fig. 6C). Altogether, these results suggested that HEFP-2b may induce arrest of cell cycle at S-phase in colon cancer cells (HCT-116). The growth-inhibitory effects of HEFP-2b correlated with their ability to arrest cells in the S-phase.

4. Conclusions

Here, a novel polysaccharide (HEFP-2b) of 3.252×10^4 Da was isolated and purified from *H. erinaceus* fruiting bodies. The polysaccharide might possess relatively stretched molecular chains and high molecular hydrophilicity and excellent molecular solubility in aqueous solution. HEFP-2b was composed of fucose, galactose, glucose, and mannose (11.81:22.82:44.28:21.09), with a backbone composed of (1→ and →6)-α-D-Glc*p*-(1→ and →4)-linked-β-D-Gal*p*-(1→ and →3,6)-α-D-Man*p* chains. Results of the MTT assay and cell cycle arrest experiments indicated that HEFP-2b significantly inhibited the growth of colon cancer cells (HCT-116) *in vitro*. The growth-inhibitory effects of HEFP-2b correlated with their ability to arrest cells in the S-phase. Our results might also provide valuable information for future studies on the use of HEFP-2b as a health-promoting functional food ingredient or therapeutic agent for colon cancer.

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CRediT authorship contribution statement

Jing-Yu Liu: Conceptualization, Methodology, Investigation, Supervision, Writing - review & editing.**Xiao-Xiao Hou:**Investigation, Writing - original draft.**Zhuo-Yu Li:**Conceptualization, Methodology.**Shu-Hua Shan:**Software, Formal analysis, Visualization.**Ming-Chang Chang:**Resources, Validation, Data curation. **Cui-Ping Feng:**Resources, Formal analysis.**Yin Wei:**Software, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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